

Feasible ML warm start for VQE

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Contents

Recommendation	2
What the literature says	2
Why not reinforcement learning ansatz search	3
Best method to implement	3
Why this method is the best fit	3
Practical design choices	4
Molecules	4
Input features	4
Model	4
Targets	4
Ansatz	4
Training labels	4
Evaluation	5
What the report can safely claim from the evidence	5
Final judgment	5
References	6

Recommendation

The most feasible four week path is a small supervised MLP that takes bond length as input and predicts a good initial VQE parameter vector, followed by a short classical refinement step. The strongest reason to choose this route is that it keeps the machine learning part simple, stays close to the chemistry use case, and is backed by two complementary strands of evidence. Geometry to parameter maps already work well on small molecules [Tao22, Cer22]. Parameter reuse across nearby geometries gives clear convergence gains even without learning, which means the target function is smooth enough to learn and the baseline is already strong [Sko23].

A full replacement of VQE optimization is riskier for a student project, even though [Tao22] shows that it can work. Reinforcement learning for ansatz search is much less feasible in the same time window because it adds a second hard problem, discrete circuit search, on top of chemistry VQE and classical training [Ost21, Kru25].

What the literature says

Small molecule chemistry already has a clear geometry to parameters pattern. In [Tao22], a feed forward network takes molecular geometry, mostly bond length in the tested cases, and predicts the nonzero ansatz parameters for UCCGSD circuits. The model uses three ReLU hidden layers and was tested on H₂, HeH⁺, H₄, LiH, and BeH₂. After training on VQE solutions from sampled geometries, the network can replace per geometry optimization and evaluate new points in one forward pass. For H₄, the paper reports about a 340 fold wall clock speedup over standard VQE because the iterative optimizer is removed.

A second line of evidence comes from direct warm starts rather than learned regressors. [Sko23] studies parameter transfer along potential energy curves with UCCSD and finds that reusing optimized parameters from a nearby geometry gives a clear gain over Hartree Fock based starts. At a transfer distance near 0.1 Å, the paper reports an average speedup of about 70 percent in the number of energy evaluations. This is especially important for the present project because it shows that nearby geometries carry useful information even before any ML is added.

[Cer22] gives a strong conceptual justification for the same choice. That paper learns a map from nuclear coordinates to circuit parameters and shows high fidelity approximate ground states across molecular geometries using only a small number of training points in several chemistry examples. The emphasis is direct state preparation rather than initialization, but the method can also serve as a warm start. The main lesson is that small chemistry systems can support low data coordinate to parameter learning when the geometry range is structured.

[Xia19] also supports the idea that bond length is an effective conditioning variable for small molecule curves in H₂ and LiH. Its model is less directly aligned with the present goal because it is a hybrid quantum classical network for energy prediction rather than a simple classical model that predicts VQE parameters. It is useful as background evidence, not as the main implementation template.

[Mes24] is closer to the desired workflow on H₂ and does use an MLP with interatomic distance as input, but it adds a quantum autoencoder, a custom angle aware loss, and a smoothed label generation pipeline. That makes it interesting as proof that the idea works, yet too elaborate for the lowest risk four week plan.

Why not reinforcement learning ansatz search

The literature on RL ansatz search is promising, but it solves a harder problem than the one that needs to be solved here.

Paper	Main task	What it improves	Cost of reproducing it
[Ost21]	RL search over circuit structure for LiH	Much shallower circuits than standard ansatzes, chemical accuracy on LiH	High. Requires RL agent, circuit environment, reward shaping, and repeated inner loop angle optimization
[Kru25]	RL search for bond dependent circuit architectures	Transferable circuits across LiH bond distances	Very high. Requires a state aware RL setup, mixed discrete and continuous actions, and substantial engineering

In [Ost21], the agent adds gates one by one and each step still needs a classical optimizer such as COBYLA or Rotosolve to tune circuit angles. In [Kru25], the setup becomes even heavier because the policy is conditioned on bond distance and outputs both gate choices and gate parameters. These papers are useful as comparison points, but they do not match the deadline or the need for a dependable demonstration.

Best method to implement

The recommended method is:

1. Sample a grid of bond lengths for H2 and LiH.
2. Run ordinary VQE at each training geometry to obtain optimized parameter vectors.
3. Train a small MLP on bond length to parameter vector.
4. At a new geometry, use the MLP prediction as the initial point for COBYLA or SPSA.
5. Measure whether the warm start reduces optimizer iterations and total energy evaluations.

This is not exactly the same as [Tao22]. That paper aims to remove the optimizer after training. The better project choice is a milder version of the same idea: use the network for initialization, then let VQE refine. This reduces risk while preserving the ML component.

Why this method is the best fit

Criterion	MLP warm start	Parameter transfer only	RL ansatz search
Includes a clear ML component	Yes	No	Yes
Close to H2 and LiH chemistry use case	Yes	Yes	Mostly LiH
Simple enough for Python and Qiskit in four weeks	Yes	Very yes	No

Criterion	MLP warm start	Parameter transfer only	RL ansatz search
Supported by direct literature precedent	Yes, [Tao22, Cer22, Mes24]	Yes, [Sko23]	Yes, [Ost21, Kru25]
Easy to benchmark against COBYLA and SPSA	Yes	Yes	Harder
Low project risk	Moderate to low	Very low	High

Parameter transfer alone is the safest engineering choice, but it does not satisfy the need for a meaningful ML component. RL ansatz search does satisfy the ML requirement, but the engineering burden is too high. The MLP warm start sits in the right middle ground.

Practical design choices

Molecules

Start with H2. Use LiH only after the H2 pipeline is stable. This follows the evidence pattern in [Tao22], [Sko23], and [Xia19], where H2 is the easiest chemistry testbed and LiH is the natural next step.

Input features

Use bond length only at first. This matches the successful small molecule setups in [Tao22] and [Mes24]. Extra handcrafted features are not needed for a first pass.

Model

Use a small feed forward MLP with two or three hidden layers. [Tao22] uses three hidden layers with ReLU and succeeds on small molecules. There is no need to match its larger widths exactly. A smaller network is enough for a course scale demonstration.

Targets

Predict the full optimized VQE parameter vector for a fixed ansatz across the whole geometry range. To make the target smoother, it is worth fixing a consistent ansatz and ordering the parameters carefully. [Mes24] shows that parameter smoothness matters a lot when training on angles.

Ansatz

Use one fixed ansatz for the learning experiment. UCCSD or a small hardware efficient ansatz are both defensible. The literature most directly tied to chemistry warm starts uses UCC family circuits [Tao22, Sko23]. A hardware efficient ansatz may be easier to manage in Qiskit if runtime becomes a concern, but the chemistry story is stronger with a UCC style ansatz.

Training labels

Generate labels offline by solving VQE on a training grid of bond lengths. This is exactly how [Tao22] constructs its dataset. For the present project, these labels can also be generated with transfer from the previous geometry so the offline stage is faster and the label sequence is smoother, borrowing a useful trick from [Sko23].

Evaluation

Compare four starts at each test geometry:

- Hartree Fock based default start
- Zero or random parameters
- Parameter transfer from the nearest solved geometry
- MLP predicted parameters

The core success metric is reduction in optimizer iterations or total energy evaluations to reach a fixed energy tolerance. This is the fairest comparison to [Sko23], which reports gains in function evaluations. Final energy error should be reported too, since a warm start that converges fast but lands in a worse minimum is not useful.

What the report can safely claim from the evidence

The literature supports three claims with high confidence.

- Geometry conditioned parameter prediction is already demonstrated for small molecule VQE style workflows [Tao22, Cer22].
- Nearby geometry information clearly helps convergence, with a reported average speedup of about 70 percent for parameter transfer at small bond length steps in [Sko23].
- Reinforcement learning ansatz search is a real research direction, but it is a poor match for a low risk four week implementation target [Ost21, Kru25].

A fourth claim should be stated more carefully. The literature strongly suggests that an MLP warm start for H₂ and LiH is feasible, but the exact reduction in iterations for this precise setup is not yet standardized across papers. [Tao22] mostly studies replacing optimization after training, while [Sko23] studies transfer without learning. The proposed project combines the most practical parts of those two lines.

Final judgment

The report converges to one recommendation: implement a bond length conditioned MLP that predicts initial VQE parameters for a fixed ansatz, then run a short VQE refinement from that starting point.

This choice is the best match to the project constraints because it is simple, clearly connected to the target molecules, grounded in existing small molecule literature, and easy to benchmark against standard initialization and against non ML parameter transfer. It is ambitious enough to be interesting, yet controlled enough to finish and present within four weeks.

References

- [Tao22] Y. Tao, X. Zeng, Y. Fan, J. Liu, Z. Li, and J. Yang, “Exploring Accurate Potential Energy Surfaces via Integrating Variational Quantum Eigensolver with Machine Learning,” *The journal of physical chemistry letters*, pp. 6420–6426, Jun. 2022, doi: [10.1021/acs.jpcclett.2c01738](https://doi.org/10.1021/acs.jpcclett.2c01738).
- [Cer22] J. Ceroni, T. F. Stetina, M. Kieferová, C. O. Marrero, J. Arrazola, and N. Wiebe, “Generating Approximate Ground States of Molecules Using Quantum Machine Learning,” Oct. 11, 2022.
- [Sko23] M. Skogh, O. Leinonen, P. Lolur, and M. Rahm, “Accelerating variational quantum eigensolver convergence using parameter transfer,” *Electronic Structure*, vol. 5, Jul. 2023, doi: [10.1088/2516-1075/ace86d](https://doi.org/10.1088/2516-1075/ace86d).
- [Ost21] M. Ostaszewski, L. M. Trenkwalder, W. Masarczyk, E. Scerri, and V. Dunjko, “Reinforcement learning for optimization of variational quantum circuit architectures,” *Neural Information Processing Systems*, pp. 18182–18194, Mar. 2021.
- [Kru25] M. Kruntunger *et al.*, “Reinforcement learning of quantum circuit architectures for molecular potential energy curves,” Nov. 20, 2025.
- [Xia19] R. Xia and S. Kais, “Hybrid Quantum-Classical Neural Network for Calculating Ground State Energies of Molecules,” *Entropy*, vol. 22, Dec. 2019, doi: [10.3390/e22080828](https://doi.org/10.3390/e22080828).
- [Mes24] K. J. Mesman, Y. Tang, M. Moller, B. Chen, and S. Feld, “NN-AE-VQE: Neural network parameter prediction on autoencoded variational quantum eigensolvers,” Nov. 23, 2024.